



Science and
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STFC Vibration Analysis Tool

CoCo-80X User Guide



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1 Abbreviations and Definitions

CSA	Configurable Signal Analysis Project: These are presets within the CoCo-80X device for viewing and analysing both live and historical data. CSAs can be customised and saved with different sampling rates, traces, input channels, analysis parameters and more depending on the specific use case.
EDM	Engineering Data Management: This software, downloadable from Product Support using the username 4202784 and password WvQxa , creates a bridge between the device and a computer allowing data to be analysed using a method of your choice. This could be the in-built analysis tool or another tool such as Matlab or Excel.
APS	Auto-Power Spectrum: The APS is the magnitude squared of the FFT of a signal, averaged over multiple blocks, showing the power at each frequency bin. For an acceleration input the units of this would be g^2 . Changing block size (N) or sampling rate (F_s) would change the APS.
PSD	Power Spectral Density: The PSD describes how the power of a signal is distributed across frequency. For a measured acceleration signal, the PSD at a given frequency tells you how much mean-square acceleration (g^2) exists per unit bandwidth (Hz) at that frequency. Unlike APS, this is independent of analyser setup (Block Size, N and Sampling Rate, F_s).
FFT	Fast Fourier Transform: An FFT allows for frequency over time data to be easily converted to a frequency distribution plot. This allows for identification of common frequencies in the system. This has both real and imaginary components, but if phase is not important in the specific use case, then the magnitude of this can be shown instead. The phase of vibrations can be useful when multiple signal traces from various positions along the beamline are compared. However, when trying to minimise the magnitude of vibrations phase is not a concern. For more information, see section 2.
FRF	Frequency Response Function: An FRF shows the response of a system to different frequencies. This is unique to each structure and is dependent on mass, stiffness and damping. an FFT is usually used to interpret this data. This is often used as a simple way of finding the resonant frequencies of a system.
DSA	Dynamic Signal Analyser: This is one of the two working modes that we have the license to use on the CoCo-80X and is the most common mode used for vibration analysis .
OMA	Operational Modal Analysis: This is the other working mode we have access to and allows us to take phase readings to complete modal analysis.
F_s	Sampling Rate: The number of data readings that are taken per second (Hz).

k	Bin Number: A number used in an iterative equation to identify what bin is active.
Δf	Frequency Resolution: The space between frequencies in an FFT.
f_n	Natural/Resonant Frequency: The peak on a FRF graph and the frequency at which excitation occurs.
ζ	Damping Ratio: The extent to which a system resists oscillatory motion.
N	Block Size: The number of samples taken per time block. A setting within the parameters of the CoCo-80X device.

2 Theory

2.1 Fast Fourier Transforms

The FFT is a computationally viable way to get to the same end result as you would using a discrete fourier transform. When using the CoCo-80X paired with an input from the accelerometer, we will have an output of voltage proportional to acceleration (g-force). This is then sampled at a fixed rate, F_s , outputting a plot of voltage over time. This is the data in its rawest form, with no analysis methods applied.

To further analyse the data, an FFT spectral analysis can be performed by the CoCo-80X. Generally, a one sided FFT will be used here which shows the absolute magnitude of the FFT, this shows all results between 0 and the Nyquist frequency. The result is a plot of amplitude (g) over frequency (Hz). It should be noted that applying just the FFT will result in a plot of amplitude (g) over k , where k is the bin number. In order to achieve the desired plot of amplitude (g) over frequency (Hz) then equation **A** must be applied. However, this is completed automatically by the CoCo-80X machine and is only shown here for reference. The Nyequist frequency is the frequency in the centre of an FFT plot and is the frequency around which a one sided FFT is taken, $F_n = \frac{F_s}{2}$, where F_n is the Nyequist frequency. It also represents a relationship between the sampling rate and the frequency of a measured signal. An example of the effectiveness of an FFT is shown in figure 1 with the transition between the time and frequency domains shown from plot 1e to 1f.

$$Frequency = \frac{k * F_s}{N} \quad (\mathbf{A})$$

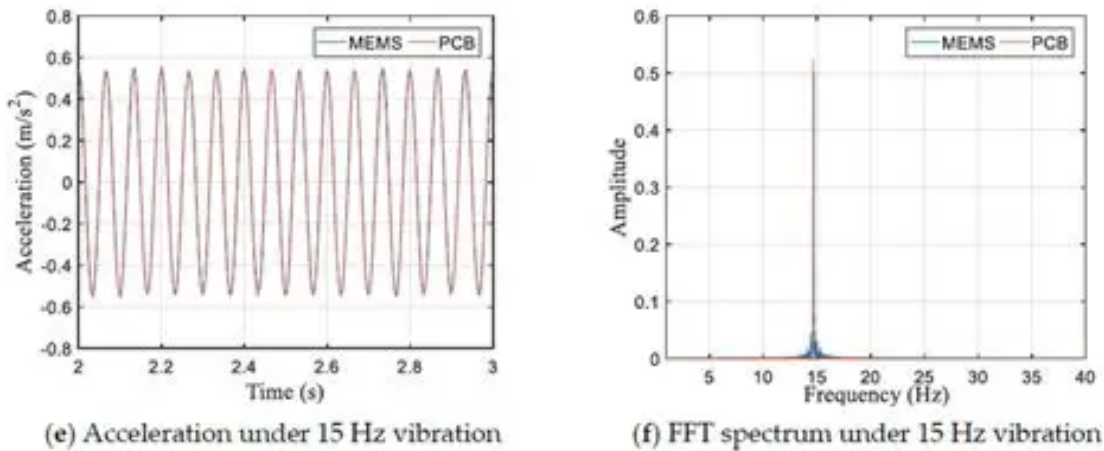


Figure 1: Accelerometer Data Transformed From the Time Domain to the Frequency Domain Using an FFT

2.2 The PSD

Once an amplitude over frequency plot has been obtained to identify vibration frequencies in the system, the data can be scaled to convert the amplitude axis into a more useful function. The most common of these is to create a PSD. This is critical to vibration analysis as it is independent of the setup features (frequency resolution) and will act as our input for an FE analysis. This is found through dividing the amplitude by the frequency resolution, giving units of g^2/Hz . The frequency resolution, Δf can be found using equation **B**.

$$\Delta f = \frac{F_s}{N} \quad (\text{B})$$

During this process the values can be converted to a physical unit rather than volts through using the calibration factor of the accelerometer. In our case this is 10 V/g. A PSD allows you to interpret more than just the resonant frequencies, as can be seen after applying the FFT transformation. From a PSD with a loglog scale, an example of which is shown in figure 2, we can find the character of the vibration source through the initial slope of the PSD. In this case, the steep slope of 40 dB/decade indicates that vibrations are most present at low frequencies as power density decreases significantly as frequency increases. A steeper slope would indicate that damping is present whereas a shallow or flat slope would indicate equal power across a range of frequencies, commonly known as white noise. Additionally, we can once again identify the resonant frequencies and from this, the damping ratio, ζ , can be estimated through equation **C**. Note Δf here is not the frequency resolution previously defined but is rather the width of the peak at $\frac{1}{\sqrt{2}}$ of the peak's maximum amplitude. f_n is the resonant, natural, frequency identified. For a detailed plot of some vibration sources you may find on a PSD, see figure 3.

$$\zeta = \frac{\Delta f}{2f_n} \quad (\text{C})$$

Other features to be aware of are sharp spikes, these are not shown in figure 2 but indicate vibrations caused by background machinery, usually rotating at a fixed frequency, such as a pump or a fan. Finally, the RMS acceleration can be found through integrating the PSD over a frequency band, this value can be compared against a specification limit and is a measure of vibration severity across the band of frequencies. The RMS acceleration associated with a single frequency band can be found through equation **D** where a_{rms} is the rms acceleration and PSD_{peak} is the value of the PSD at the frequency of interest. Note that the size of the frequency band in Hz is equivalent to Δf . The peak acceleration across the frequency band, rather than the RMS, can be found through equation **E**.

$$a_{rms} = \sqrt{PSD_{peak} * \Delta f} \quad (\text{D})$$

$$a_{peak} = a_{rms} * \sqrt{2} \quad (\text{E})$$

Whilst the maximal displacement at any frequency can be found using the EDM software for our use case with the CoCo device, this can also be found through using equation **F**.

It is key here to use the value for RMS acceleration and not the peak acceleration. In this case, s is the displacement and f is the target frequency used to find the PSD value when calculating the RMS acceleration.

$$s = \frac{a_{rms}}{(2\pi f)^2} \quad (\mathbf{F})$$

Importantly, when two PSDs are compared, for example a PSD of an accelerometer on the ground and an accelerometer on a table, the ratio between them indicates how the structure modifies the ground motion and what frequencies are amplified by this. This is called transmissibility (T)

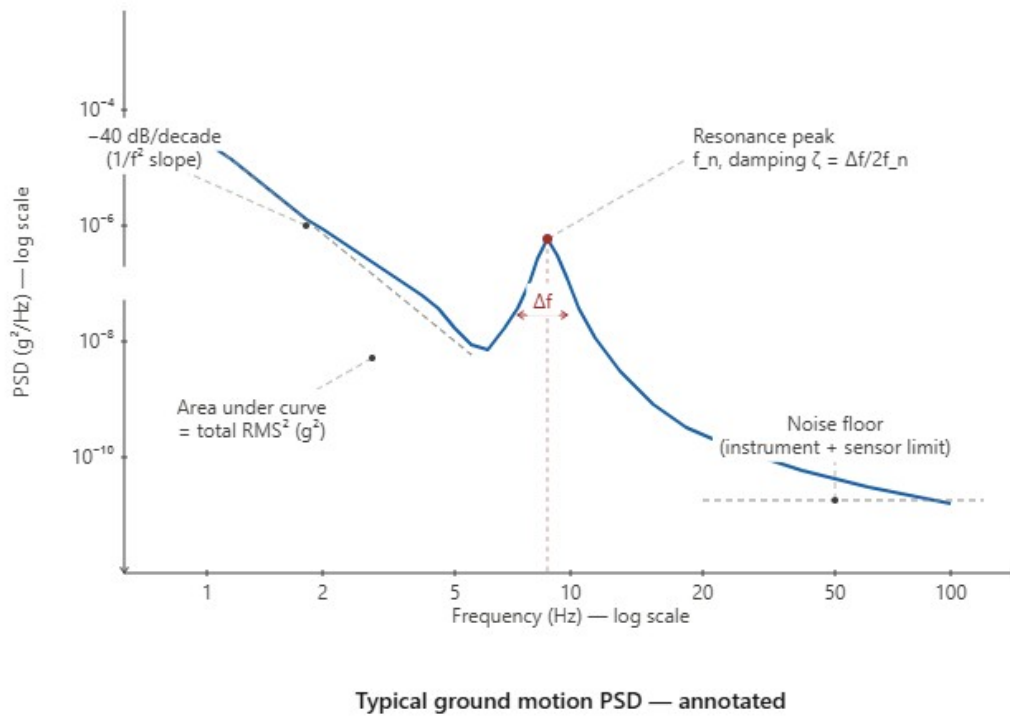


Figure 2: Annotated Example PSD Plot on a LogLog Scale

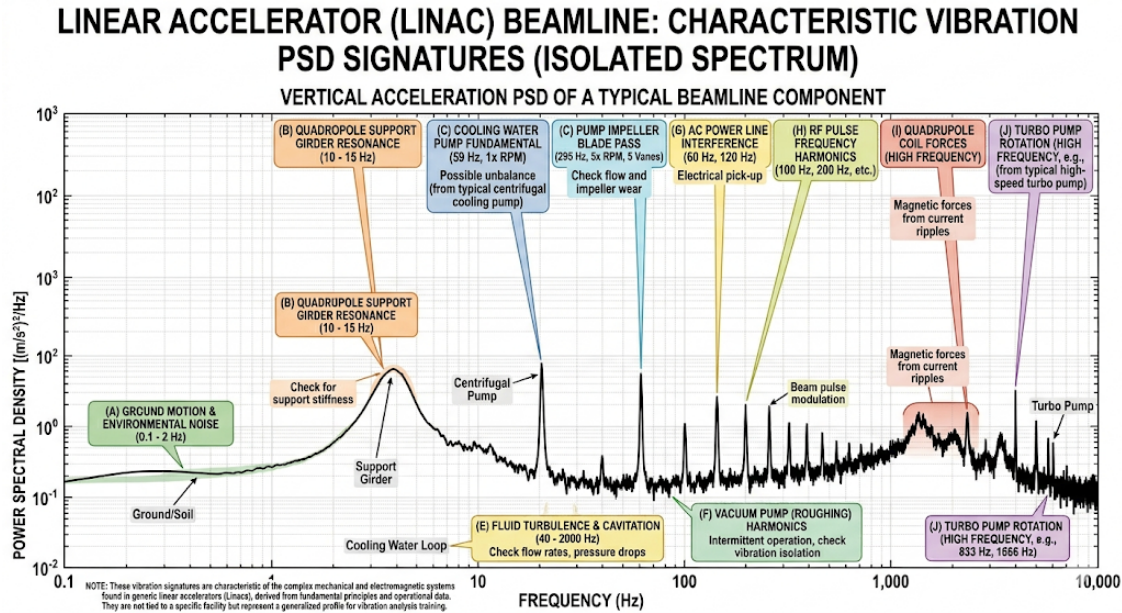


Figure 3: Examples of Vibration Sources on a PSD

3 Device Setup

Upon switching the CoCo-80X device on, the user should wait 20 minutes for the device to 'warm up'. This will allow for the most accurate measurements. The user will then be presented with one of two pages. The first option will be the selection of the working mode, in our case the choice between DSA or OMA mode. For the majority of applications, DSA mode will be the correct selection here. OMA is present for the application of measuring phase difference where this is required, however, this is something that can also be analysed after exporting the data to another analysis software such as Matlab. Alternatively the user will be presented with a page of recent projects. If this is the case then it is recommended to press f1 followed by f6 and select the required mode, the user should then be presented with the recent projects page again. This page can always be returned to from the home page through using the return button on the keypad. Note all buttons on the keypad are labelled in figure 4.

Following this, a recent project can either be selected to work on or the user can return to the home page using f1 where the two most common actions would be to view previously recorded events through pressing the files button on the keypad or to begin a new analysis through selecting the CSA Application. In the CSA Application the user can choose from a large range of CSAs, which are essentially analysis presets, based on the user's use case. Ideally, once sufficient testing has been completed, the STFC users will set up their own standard CSAs for their most common use cases as a time saving procedure. This can be done most easily through editing a CSA already on the device. CSA presets are sorted into a number of categories, the most frequently used of which are transient and spectral analysis, and frequency response. With the STFC license the user will have access to all groups down to vibration intensity. All CSAs can be viewed through pressing f3 and, once set up using

Button Name	Functions
 Power	Power on/off the CoCo-80X or lock keypad
 Save	Save the signals designated by the CSA project
 Setup	Go to the main setup page
 Analysis	Open the “Analysis Groups” page to choose a CSA project
 Display	Switch to the current CSA project signal display (must have an active project running)
 File	View existing record files stored on CoCo memory.
 Input	Enter the input channel setup page
 Trace Selection	Go to previous or next trace window
 Recording	Start/Stop recording time stream(s)
 Save	Save signal(s)

Figure 4: CoCo-80X Summary of Keypad Buttons

the EDM, custom CSAs can be viewed using f2. To record an FFT and therefore a PSD, a CSA from the transient and spectral analysis section should be used whereas the frequency response section can calculate an FRF.

The number in brackets next to the CSA denotes the number of channels enabled, whilst a single channel will allow you to view ground vibrations it is likely that at least 2 channels should be enabled to view differences between ground and system vibration and therefore examine the vibration dynamics of the system. Through enabling multiple channels, raw data and integrated data (through an APS or PSD graph) can be displayed simultaneously. A PSD can be displayed through pressing the top left hand corner of a trace and selecting the unit $\frac{EU^2}{Hz}$. The scale of a trace can also be manually set through pressing on it.

Once a CSA has been selected, the pre-selected graphs should be visible. Without an input connection, such as an accelerometer, placeholder data will be shown in the graphs. Ignoring this placeholder data, the user should connect an input, most likely an accelerometer, to channel 1 followed by channel 2 and so on. To do this, go to the CSA page, if on the home page then this can be done through pressing the display button on the keypad. The user should then press f2 followed by input channels. The user must then customise the

input parameters based upon the input devices specification, for example the calibration factor of the accelerometer. This can be done through selecting one of the accelerometers that an STFC staff member has input into the device previously (through identifying the accelerometers part number or serial number). Alternatively, the user could input their own information. The key parameters to change here would be the measurement quantity, likely changed to acceleration for our use case and the sensor sensitivity, 10000 mV/g for the most modern of our accelerometers. This should then be applied to all channels using f1 and the sensor sensitivity can be adjusted across channels dependent on the input device parameters. When using the PCB triaxial 356B18 accelerometer, the required sensor sensitivity will be 1000 mV/g rather than 10000. As a standard, channel 1 should be used for the ground accelerometer and, if using a triaxial accelerometer; the x, y and z axes should be in channels 2,3 and 4 respectively. If using a uniaxial accelerometer on the target site then this can use channel 2 and channels 3-4 can be removed from the spectral save. When changing accelerometers, ensure that the sensor sensitivity is correct in the input settings.

An input mode for each channel will also have to be selected. Most of the time, the IEPE mode is recommended, especially when no common mode voltage exists like in the case of measuring the output of sensor amplifiers. If a common mode voltage does exist then this will create noise in this mode.

For particle accelerator ground motion studies, very low sub-Hz frequencies are of importance. For this use case, it is important to record data for a long time, therefore achieving a large value of N, to get a low frequency resolution. Here, a long time would be considered approximately 5 minutes. Repeatability is also important and so it would be worth taking multiple 2-3 minute recordings at different times rather than a single 5 minute recording as this will result in PSDs that can be compared. The CoCo-80X can record data at a sampling rate of up to 102.4 kHz. However, this should be chosen based on the specific use case. We know through equation **B** that the sampling rate is directly proportional to the frequency resolution, therefore a low sampling rate is desirable for our use case, with the trade off that higher frequencies will be undetectable. Frequencies of up to half the sampling rate are detectable, this maximum frequency is the Nyquist frequency. For measuring vibrations on an accelerator beamline, a **sampling rate between 1kHz and 2kHz** would be recommended. As we know, to detect low frequencies accurately a low (fine) frequency resolution is desirable, decreasing the gap between bins without having to significantly increase the recording time. However, for very short, transient events that may only last a number of milliseconds a high sampling rate is required to capture this. Therefore, a balance must be struck to ensure accurate recording of data without unreasonably increasing the required recording time. The sampling rate can be altered through returning to the trace page of the CSA and then pressing f2 and selecting 'sampling rate (Fs)'.

To complete the initial setup, press f2 and select 'analysis parameters' on the CSA trace page. Here, the key parameters to set would be **block size** and **average mode**. As we know from section 2, the block size is the number of samples that are taken. The second number after the forward-slash on the CoCo-80X is the number of spectral lines (data points) in the frequency domain of the FFT. Increasing the block size decreases the frequency resolution and allows

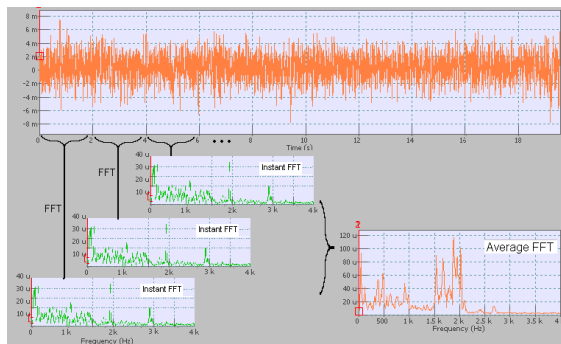
detection of lower frequencies. However, it also increases the calculation time and response time. Due to RAM and processing constraints within the CoCo-80X, a large block size will decrease the maximal possible recording time when using the desired linear averaging mode. For our use case, to achieve a suitably low frequency resolution whilst maintaining a low processing requirement, a **block size of either 2048 or 4096** should be selected. This will allow for at least a minute of recording using the linear averaging mode. The overlap ratio and average mode options can be adjusted to optimize the FFT operation for specific signal characteristics. When choosing an average mode, the exponential mode should be used to weigh the most recent FFT higher, linear will compute X amount of FFTs and then stop, peak hold will maintain the highest value for each frequency, time linear will average X amount of time blocks and then calculate the FFT and time exponential is a running average of the blocks, weighing the most recent more heavily. A visual representation of time vs frequency domain averaging is shown in figure 5. Frequency averaging will not show phase data so should only be used where magnitude of vibrations is the focus. Time averaging will show phase and is useful for filtering out unwanted noise when trying to detect a specific vibration source. However, a trigger for block synchronisation is required to use this mode which can be added in the trigger settings (f2 after selecting a CSA). The exponential average, as we know, will compute older spectra with a lower weighting. This is useful when looking at slowly evolving phenomena that require a long recording time. With the linear averaging mode, the recording time that the CoCo-80X is capable of is relatively short (a few minutes) whereas with the exponential mode this limitation is not present. The peak hold average will only be used when looking at the worst case scenario for each frequency rather than providing a complete picture of the vibration mechanics of a system. It is likely that for our use case the linear averaging method will be the best to use. f6 should then be used to apply the changes made. A large overlap ratio will allow for increased accuracy of data, as this determines how much blocks overlap with each other. However, this also increasing the processing requirement as more FFTs are being computed per second and so for this reason a **50% overlap ratio** has been selected for our use.

Once the initial setup has been completed, it is key to select the correct traces. This was likely completed when selecting a CSA but in the case that the user would like to select different traces they should press f1 followed by 'trace and window settings'. Here the user can choose from preset traces or can create their own through pressing f1 and selecting their desired option. Whilst this allows the user to display the desired traces, it does not dictate what traces are saved, more on this in section 4. To set the traces that are saved in both the time and frequency domain the user should press f2 followed by spectral save setup. The available options will depend on what traces have been selected in the trace and window settings however it is likely that not much will have to be changed if using a preset CSA made by either crystal instruments or an STFC staff member. The desired output channels should be selected and the user can choose to either save block signals continuously or save signals every 5 seconds. For optimal results it is recommended to select save block signals continuously.

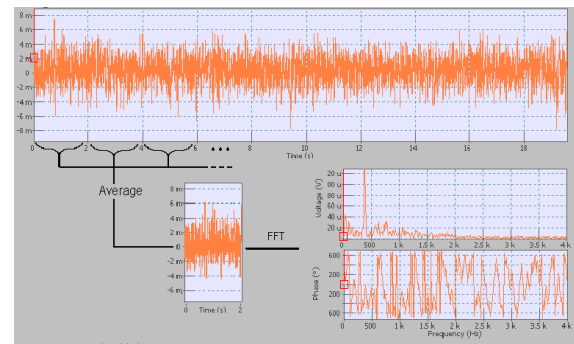
Mode	Domain	Description	Typical Use Case
Linear	Frequency	Computes a fixed number (N) of FFT spectra and averages them with equal weighting, then stops.	Stable, stationary signals where a clean averaged spectrum is required over a defined measurement window.
Exponential	Frequency	Continuously averages FFT spectra, applying exponentially decreasing weights to older frames so that more recent spectra contribute more to the result.	Slowly evolving or quasi-stationary signals where continuous spectral monitoring is needed without a fixed stop point.
Peak Hold	Frequency	Retains the maximum amplitude recorded at each frequency bin across all computed FFT frames; does not average.	Worst-case vibration assessment — identifying the highest energy ever observed at each frequency rather than a representative average.
Time Linear	Time	Averages a fixed number (N) of time-domain blocks before computing the FFT, then stops. Preserves phase information.	Extraction of a coherent signal from a noisy measurement when a trigger is available for block synchronisation.
Time Exponential	Time	Applies exponentially decreasing weights to successive time-domain blocks before computing the FFT, providing a running average. Preserves phase information.	Continuous monitoring of a phase-coherent signal source in the presence of noise, using a synchronisation trigger.

Note: Frequency-domain averaging modes (Linear, Exponential, Peak Hold) discard phase information and are suited to magnitude-only vibration analysis. Time-domain averaging modes (Time Linear, Time Exponential) preserve phase and require a trigger for block synchronisation, configured via the trigger settings menu.

Table 1: Summary of averaging modes available for spectral analysis.



((a)) FFT Averaged in the Frequency Domain



((b)) FFT Averaged in the Time Domain

Figure 5: A Comparison of FFT Averaging Modes

4 Data Analysis

The device is now setup and ready for data analysis. It is likely that the user will want to record data to refer to at a later date. To do this, press the save button on the keypad and information about the recording will appear at the top of the screen. This includes the recording time and file size. To stop the save, press the same button. It is imperative that the user does not use the record function as this will only save time stream data, with no information on the FFT. For a guide on how to use an experimental feature allowing the user to record and save at the same time, see the dedicated folder within PME, Engineering Information, Vibration, 2025 New Vibration Handheld.

There are a number of options for data analysis. To view data live the user can either examine data on the CoCo-80X traces or they can view a live transmission of the screen from a connected workstation within the EDM software. This would allow the user to view data without the possibility of changing the recorded data through knocking a sensor, for example. Alternatively, once the recording has been completed, the included USB drive can be used to take the data from the device and view graphs on the EDM software after the fact. Note that if the data was not recorded then this will not be possible. Through using the EDM approach, alternative methods such as Matlab or Excel could be used to analyse data further. For STFC applications, it will likely be necessary to record raw data (acceleration) and the FFT. This will allow us to find a PSD during post processing. For best results, it is recommended to use the EDM software to view data after saving has finished. To do this, the CoCo-80X must be connected to the PC following the below steps.

To remotely connect to the CoCo-80X the user must first download the EDM software through the link and login details provided in section 1. The user should download the bottom option, engineering data management, through which they can download all relevant applications, the most important being the EDM. When prompted for a license key during EDM setup, the user must go to PME, Engineering Information, Vibration, 2025 New Vibration Handheld folder and select the .lic file. The CoCo-80X device must then be connected to a network wide ethernet port on the laboratory campus, specifically for use at a UKRI lab. Provided this ethernet port is connected to the same network as the user's PC, the user should be able to connect to the device through opening the EDM software and clicking: project, add CoCo, add using IP address, 148.79.103.1, add. Remote viewer can be enabled on the corresponding tab within the EDM app and the user should then be able to use the CoCo-80X as if they were on the device itself.

The recommended way to analyse data would be to view traces from a recording after they have finished recording, rather than live. To do this, download the desired recording through the file viewer tab and then right clicking on the file and selecting 'download'. Following this, the file can be seen in the CoCoData Files tab in the bottom right of the EDM, as seen in figure 6. The user should double click on the file and open up both time and frequency signal folders, dragging the desired traces within this folder into the main trace viewer. To view more traces as seen in the example, right click on the name of the signal display tab, for example fftwohalf_BLOCK(ch1) and add the desired tabs through using the table.

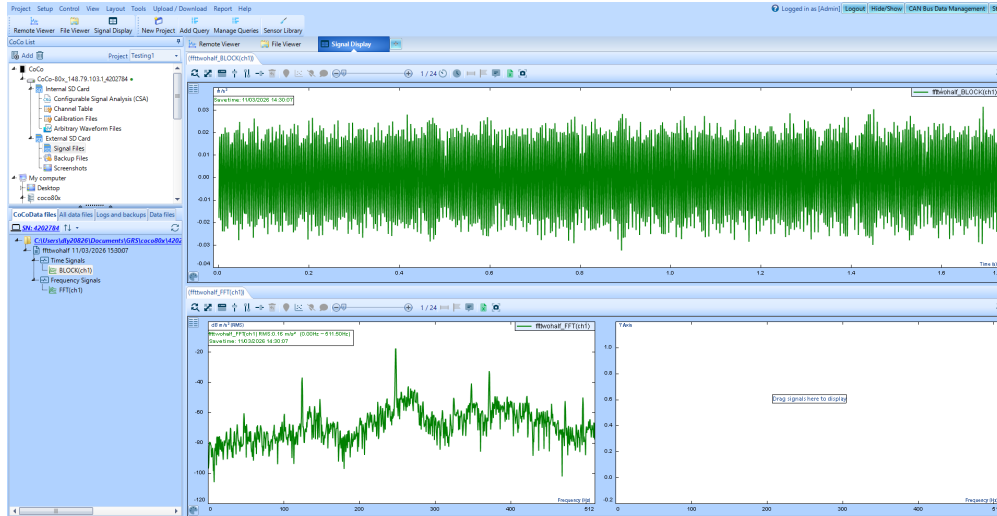


Figure 6: Example of Analysing Traces on the EDM Software

The user can then add more traces by right clicking on the tab name and selecting the amount of trace boxes they want. Note that each time trace blocks are made these block will be related to each other and so all traces on this section must have the same domain (x-axis) of either time or frequency. Additional signal display tabs can be opened by clicking the plus icon next to the existing signal display tab. It is common that the user will want to convert the FFT to a PSD. To do this first right click on the trace and change y axis format to linear mag (magnitude) and then change the spectrum type to $\frac{(EU)^2}{Hz}$ where EU is the unit of the data. On the time domain data, displacement can be viewed through right clicking and changing the option acc/vel/disp to displacement. It should be noted that this is not a recommended way of analysing displacement due to the double integration causes errors of up to an order of magnitude. As previously mentioned, the user may want to export data to excel for analysis. To do this, right click on the trace file in CoCoData Files (in the bottom left of the EDM screen) and select 'Export Data To Excel' or 'Export Data To File'. Following this, acceleration and displacement PSDs can be found through using the python code provided in the CoCo-80X Folder. To do this you will first need to download an IDE for example Spyder or VSCode. If using VSCode, the python extension will have to be installed and then the user should type **pip install numpy** in to the terminal to install additional extensions that are required.

5 Ansys FEA for Vibration Analysis

When performing vibrational analysis on a CAD model in Ansys FEA, there are three primary modes to choose from. These are: Transient Structural, Random Vibration and Harmonic Response. Each of these will require a different data output from the CoCo-80X. For transient simulations the user will need the raw data in g or m/s^2 . For random vibration simulations the user will need a PSD exported to a csv. For harmonic response simulations an FFT input is required. These three methods can be used to find natural frequencies and

predict displacement throughout the system as a result of vibrations. This displacement will be significantly more accurate than that found through integration of acceleration data. The first data entry for inputs into ansys must always be at $t=0$ to avoid a simulation failure.

6 Data Management Standards

To avoid confusion when viewing and analysing data, it is important to set standards for naming conventions, coordinate systems and file storage locations in addition to defining a post-recording routine to provide clarity for all users of the data.

Firstly, through the spectral save menu, trace saves can be given a name. This name should begin with a location or an abbreviation for this location followed by a short description of the operating conditions. An example of this would be lr-pumpon for a reading taken in the laser room with the pump on. This name can be described in a post save document explained later. When the files have been downloaded onto a PC for data analysis, they should be moved to shared PME folder system specific to the project they are relevant on and then deleted from the CoCo device. This ensures that the SD card does not fill up and that future users do not accidentally download old data.

Perhaps most important is the writing of a post-recording document. This document is important for anyone attempting to understand the data in the future and describes the coordinate system, possibly through images, and the naming convention used for the data files. As a convention, the coordinate system should use the z axis as up and the x axis in the beamline direction when analysing beamlines. As previously mentioned, x, y and z should be connected to channels 2, 3 and 4 respectively.